ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Expanding Brackets

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

EXPANDING BRACKETS · CH2 · L3

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The Map

This lesson collapses Expanding Brackets into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Expand brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 20 website questions, 20 checked solutions, 3 local PDFs read.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q12

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

Finish: (a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 12 (a) Express 4 2 3 1 x x - - + as a single fraction. Give your answer in its simplest form. (3) (b) Expand and simplify 2x(x - 5)(x - 3) (3)

YOUR SOLUTION

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(64p^9q^{12})^{2/3} = 16p^6q^8$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P2H Q12

(a)

$$(64p^9q^{12})^{2/3} = 16p^6q^8$$

(b)

$$\frac{2}{3x} + \frac{4}{5x} - \frac{9}{10x} = \frac{20 + 24 - 27}{30x} = \frac{17}{30x}$$

Finish: (a) $16p^6q^8$, (b) $\frac{17}{30x}$, (c) $8x^3 - 28x^2 - 60x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 12 (a) Simplify (64p9q12) 2 3 (2) (b) Write as a single fraction 2 3 4 5 9 10 x x x + - Give your answer in its simplest form. (2) (c) Expand and simplify 4x(x - 5)(2x + 3) Show your working clearly. (3)

YOUR SOLUTION

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q1

Expand brackets

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

Finish: 28, $2p(4p-1)$, $12t^2 - 8t$, $5x^2 + 18x - 8$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 1 w = 5y 2 - y 3 (a) Work out the value of w when y = -2 w = (2) (b) Factorise fully 8p 2 - 2p (2) (c) Expand 4t(3t - 2) (2) (d) Expand and simplify (5x - 2)(x + 4) (2)

YOUR SOLUTION

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q12

(a)

$$\frac{4}{x-2} - \frac{3}{x+1} = \frac{4(x+1) - 3(x-2)}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3) = 2x(x^2 - 8x + 15)$$

Finish: (a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3 - 16x^2 + 30x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 12 (a) Express 4 2 3 1 x x - - + as a single fraction. Give your answer in its simplest form. (3) (b) Expand and simplify 2x(x - 5)(x - 3) (3)

YOUR SOLUTION

05 Expand brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q1

Expand brackets

$$w = 5(-2)^2 - (-2)^3 = 20 + 8 = 28$$

Finish: 28, $2p(4p-1)$, $12t^2 - 8t$, $5x^2 + 18x - 8$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 1 w = 5y 2 - y 3 (a) Work out the value of w when y = -2 w = (2) (b) Factorise fully 8p 2 - 2p (2) (c) Expand 4t(3t - 2) (2) (d) Expand and simplify (5x - 2)(x + 4) (2)

YOUR SOLUTION

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Expanding Brackets question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4x^2 - 25y^2 = (2x - 5y)(2x + 5y)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P2HR Q13

(a)

$$4x^2 - 25y^2 = (2x - 5y)(2x + 5y)$$

(b)

$$4x(x + 3)(2x - 5)$$

Finish: (a) $(2x - 5y)(2x + 5y)$, (b) $8x^3 + 4x^2 - 60x$

FORMULA BANK

Every term in one bracket multiplies every term in the other bracket.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 13 (a) Factorise 4 25 2 2 x y - (2) (b) Show that 4 3 2 5 x x x ( )( ) can be written in the form ax bx cx 3 2 + + where a, b and c are integers to be found. (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the target
"work out", "show", "hence", a familiar exam phrase	Write the first line
"work out", "show", "hence", a familiar exam phrase	Calculate carefully
"work out", "show", "hence", a familiar exam phrase	Answer in the requested form
"work out", "show", "hence", a familiar exam phrase	Expand brackets
"work out", "show", "hence", a familiar exam phrase	Factorise expression

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.